How a network learns

AI · Level 3 · Lesson 2



Today we will

- Explain how layers of neurons combine straight lines into shapes one neuron can’t learn.
- Describe training as a loop: guess, measure the error, nudge every weight a little downhill, repeat.
- Explain why the step size and enough varied examples matter.

Remember

- Each hidden neuron draws a line; the next layer combines the lines into shapes, like a band or a ring.
- Training is gradient descent: guess, measure the error, nudge every weight downhill, repeat.
- Steps that are too big never settle; too small is slow. Varied examples teach the real pattern.

Warmer, colder

1. Warm-up (5 min): One neuron, one line: recap with the umbrella.
2. Layers (8 min): Two lines fence in “just right”.
3. Warmer, colder (12 min): Find the hidden chalk by feedback alone.
4. Big steps, small steps (8 min): Replay with giant, then tiny steps.
5. Wrap-up (7 min): Map the game onto the training loop.

Choose a hiding place that needs about ten normal steps to reach. Agree the rule: the class says one word after every step, and only “warmer” or “colder”.

Let’s discuss

1. In the game, what was the error, what told you which way to go, and what was a step?
2. Why did giant steps make it harder to find the chalk?
3. Which examples would a network need to work well in our town?

Exit ticket

- Write the four steps of the training loop, and one reason the step size matters.